**Inorganic Cumulative exam**
**12-1-2017**

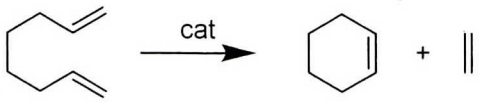
Answer five of the following questions. Only the first five of your answers will be graded. Each Question is worth 20 pts.

## Organometallic Chemistry

1. Describe metal carbonyl back bonding and provide a concise description of the orbitals involved. Describe a method you could use to quantify this phenomenon. You may use any combination of diagrams and text you choose.

2. Describe the orbitals involved in the oxidative addition of dihydrogen to Vaska's Complex.

3. Provide a catalyst and complete mechanism for the following reaction




4. Provide the mechanism for the hydrogenation of ethylene by Wilkinson's catalyst ( $RhCl(PPh_3)_3$ )

5. Describe the origin of the "18 electron rule" for octahedral organometallic complexes.

## Solid State and Materials Chemistry

6. A Gouy balance is used to quantify the magnetic susceptibility of solids and materials. Describe how this device measures magnetic susceptibility. You may use any combination of written descriptions or diagrams in your answer.

7. Graphite and diamond are two allotropes of pure carbon. One of these is a conductor the other is an insulator. Describe the differences between these two allotropes and how those differences give rise to the vastly different electronic properties of these solids

8. Define the photoelectric effect and how it was discovered

9. Why do optical band gaps typically increase with decreasing size of nanomaterials?

10. Describe the difference between a perovskite and rutile crystal lattice. Give an example of each.